เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 2

Year: 2014

Subject: 392132 Physics II

Section: 15 - 16

Date: 13 March 2015

Time: 13.00 - 15.00

Name________________________ ID ____________________ Class____________

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken inside the examination room.
3. Textbooks are **NOT** allowed.
4. Express your answer in English. Other languages will be discarded.
5. A scientific calculator is allowed.
6. Electronic communication devices are **NOT** allowed in the examination room.
7. The examination has 6 pages (including this page), 10 questions and a total score of 35 points.
8. Write all your answers on this exam sheet.

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1. A ball is thrown upward from the top of a building at an angle of 37.0° to the horizontal and with an initial speed of 30.0 m/s. The point of release is 45.0 m above the ground. Neglect air resistance.

(a) Maximum high above the ground (1 point)

(b) Time at maximum high (1 point)

(c) Velocity at maximum high (1 point)

(d) How long does it take for the ball to hit the ground? (2 points)

(e) Find the ball's speed at impact. (2 points)

(f) Find the horizontal range of the ball. (1 point)

2. A tennis player hits the 144-gram ball perfectly horizontally at a speed of 50 km/h. When it leaves the racquet the ball is 1.5 m above the ground. How far from the tennis player will the ball hit the ground ? (2 points)

3. During training, an aerial skier takes off from a ramp that is inclined at 40.0° to the horizontal and lands in a pool that is 10.0 m below the end of the ramp. If she takes 1.50 s to reach the highest point of her trajectory, calculate.

(a) the speed at which she leaves the ramp (1 point)

(b) the maximum height above the end of the ramp that she reaches (2 points)

(c) the time for which she is in mid-air. (1 point)

4. (a) What is the tangential acceleration of a bug on the rim of a 20.0-cm.-diameter disk if the disk accelerates uniformly from rest to an angular speed 75.0 rev/min in 4.00 s? (2 points)

(b) When the disk is at its final speed, what is the tangential velocity of the bug? (1 point)

One second after the bug starts from rest, what are its (answer question (c) to (e)

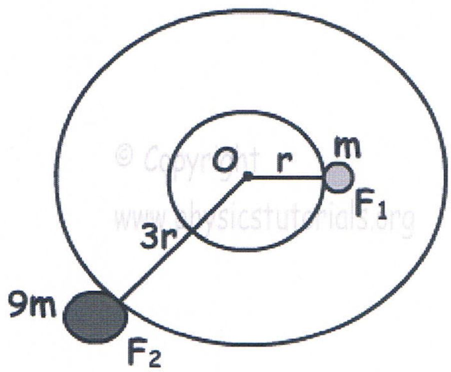
(c) tangential acceleration (1 point)

(d) centripetal acceleration (2 points)

(e) total acceleration? (1 point)

5. Coefficient of static friction of a circular road, having radius 50 m, is 0.2. Find the initial velocity of the car having 1500 kg to turn junction safely. (2 points)

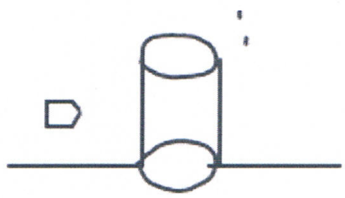
6. Two objects rotate with same frequency around point O. One of the objects has mass m and other one has mass 9m. If centripetal forces exerted on objects having mass m is F_1, and object having mass 9m is F_2, find ratio of F_1/F_2. (3 points)



7. Sheila has just arrived at the airport and is dragging her suitcase to the luggage check-in desk. She pulls on the strap with a force of 190 N at an angle of 35° to the horizontal to displace it 45 m to the desk. Determine the work done by Sheila on the suitcase. (2 points)

8. A 7.00-kg bowling ball moves at 5.00 m/s. How fast must a 2.45-g Ping-Pong ball move so that the two balls have the same kinetic energy? (2 points)

9. A 7.80-g bullet moving at 450 m/s penetrates a tree trunk to a depth of 4.0 cm.
(a) Use work and energy considerations to find the average frictional force that stops the bullet. (2 points)
(b) Assuming the frictional force is constant, determine how much time elapses between the moment the bullet enters the tree and the moment it stops moving. (2 points)



10. A 0.300-kg block along a horizontal track has a speed of 3.0 m/s immediately before colliding with a light spring of force constant 6.50 N/m located at the end of the track. What is the spring's maximum compression if the track is frictionless? (3 points)

Good Luck

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